

HARIHARAN SELVAM

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Summary

ML Engineer with a strong foundation in predictive modeling and scalable ML systems. Proficient in deploying containerized applications and optimizing pipelines for real-time analytics. Demonstrated ability to integrate robust ML frameworks with API and data-driven architectures to deliver efficient, high-performance solutions.

Education

University of Texas at Arlington <i>M.S., Applied Statistics and Data Science (GPA: 3.9/4.0)</i>	Aug 2023 - May 2025
Anna University <i>B.E., Electronics and Communication Engineering (GPA: 8.4/10)</i>	Aug 2019 - Apr 2023

Professional Experience

University of Texas at Arlington <i>Research Data Scientist</i>	Jul 2025 - Present <i>Arlington, TX, USA</i>
- Engineered a Python-based NLP pipeline leveraging SpaCy and DistilBERT to classify Medicaid waiver documents, elevating accuracy to 88% and reducing manual review time by 45%. - Conducted exploratory data analysis and statistical validations on unstructured policy datasets, generating reproducible insights to support robust ML systems. - Developed Power BI dashboards to visualize model metrics, enhancing transparency for research and facilitating data-driven decisions in ML project contexts.	
Vibhathi Labs Inc <i>Data Engineer Intern</i>	May 2024 - Aug 2024 <i>Wilton, CT, USA</i>
- Architected ETL pipelines in Python and SQL to process 1M+ structured and semi-structured records, ensuring scalable and maintainable data integration. - Optimized SQL queries, reducing runtimes by 25% and reinforcing reliable data provisioning for downstream ML applications. - Designed predictive and descriptive dashboards using Power BI and Tableau to monitor KPIs, reducing reporting turnaround by 40% and supporting ML-driven operational insights.	
Combat Vehicles Research and Development Establishment (CVRDE) <i>Data Analyst Intern</i>	Jul 2022 - Oct 2022 <i>Chennai, TN, IND</i>
- Analyzed sensor and telemetry data from combat vehicle subsystems and applied Python-based denoising techniques to enhance signal quality by 28%, supporting ML model input reliability. - Constructed predictive maintenance models for ECUs that achieved 85% accuracy, facilitating performance optimization and ML integration. - Automated pipelines for test benches, reducing manual logging by 45% and improving the accuracy of system performance reports.	

Projects

Bank Marketing Campaign Optimization	2024
- Consolidated 200K+ customer records via ETL into relational models; performed EDA to identify campaign success factors. - Built predictive classifiers (Logistic Regression, Random Forest), improving targeting effectiveness by 18%. - Tuned thresholds with ROC/AUC analysis to balance false positives/negatives, strengthening lead prioritization. - Developed automated KPI dashboards in Power BI with DAX alerts, enabling stakeholders to track campaign ROI in real time. - Collaborated with cross-functional stakeholders to document data mappings and validation protocols, improving transparency and adoption.	
Online Shoppers' Purchase Analysis	2024
- Processed and analyzed 150K+ e-commerce sessions; predictive models (SVM, Logistic Regression) achieved 85% precision and 82% recall. - Engineered behavioral features and applied PCA to reduce dimensionality while retaining 95% of variance. - Automated analytics pipeline integrated with Power BI, providing real-time insights into customer conversion funnels. - Generated SQL-based segmentation reports by channel, region, and time-of-day, guiding targeted marketing strategies. - Conducted correlation analysis between website usability factors and purchase behavior, surfacing 3 actionable insights that boosted retention by 15%.	
Potato Disease Detection Model	2024
- Trained CNN model on 50K+ labeled plant leaf images, achieving 90% test accuracy for disease classification. - Deployed model into a Flask API, enabling real-time inference and visual feedback for end-users. - Applied Grad-CAM explainability techniques to visualize CNN activations, improving trust in predictions. - Performed hyperparameter tuning and cross-validation, reducing classification error variance by 12%.	

Technical Skills

Programming: Python, R, SQL, Bash, Kotlin, Rust
Libraries/ML: Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow, SpaCy, NLTK, Transformers (HF), Machine Learning
Data Engineering: ETL Development, Data Cleaning, Databricks, Spark, Hive
Visualization: Power BI, Tableau, Matplotlib, Seaborn, Excel
Databases: MySQL, PostgreSQL, Oracle, BigQuery
Other: Git/GitHub, REST APIs, Docker, Agile/Scrum, GraphQL, Kubernetes, CI/CD, Cloud Platforms, Analytical Thinking